FEA of Sustainable Helmet Materials

under EN 397 Impact

Your Name

April 2026

ECM3175 Final Year BEng Mechanical Engineering
University of Exeter

The Problem Context

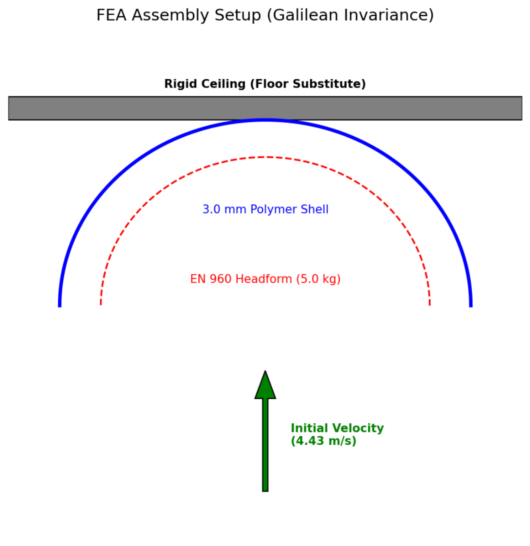
- **Environmental Impact:** Standard PPE (e.g., Virgin HDPE) has a high environmental cost.
- **The Challenge:** Transitioning to sustainable alternatives without compromising safety.
- **Regulatory Standard:** EN 397 is the safety-critical regulation for industrial helmets.

Material Candidates

Material	**Modulus (GPa)**	**Yield Stress** (kg/mm^2)
Virgin HDPE (Baseline)	1.00	2.55
ABS Plastic	2.10	4.08
Bamboo-HDPE (30% wt)	1.50	3.26
Recycled PC (rPC)	2.30	6.12

Methodology & Impact Setup

- **Setup:** Galilean upward velocity shift applied to helmet/headform.
- **Velocity:** 4.43 m/s against a rigid ceiling.
- **Solver:** LS-DYNA Explicit Dynamics.



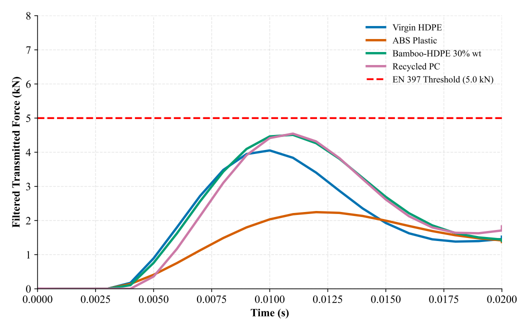
Computational Rigor & Stability

- **Time Integration:** Explicit Central Difference Method.
- **Stability:** CFL condition ($\Delta t \leq L_{min}/c$).
- **Validation:** 0.00 J hourglass energy, $< 0.04\%$ energy drift.
- **Mesh:** Convergence achieved at 2 mm (42,000 elements).

PLACEHOLDER FEA Mesh (2mm elements at crown)

Phase 2: EN 397 Force Transmission

- **Criterion:** Transmitted force must be $< 5.0\,\text{kN}$.
- **Results:** All materials pass.
  - Virgin HDPE: 4.05 kN
  - ABS: 2.25 kN
  - Bamboo-HDPE: 4.51 kN
  - Recycled PC: 4.54 kN
- **Signal Processing:** Data filtered using SAE J211 CFC 60 to isolate macroscopic deceleration.

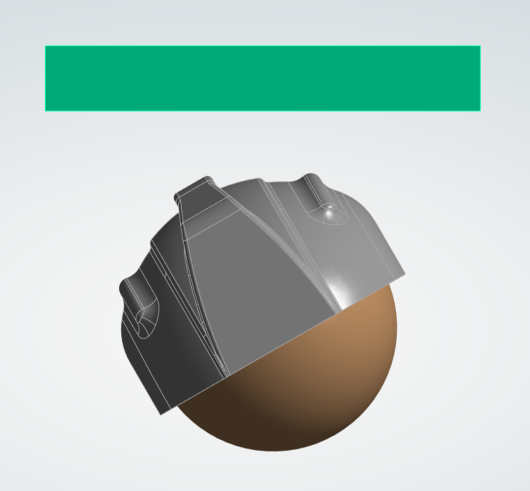


Phases 3 & 4: Energy Absorption Mechanisms

- **High Ductility (Virgin HDPE):**
  - Absorbs impact energy primarily through internal strain yielding and plastic deformation.
  - Energy Absorption Ratio: 55.3%.
  - Crush Depth: 18.42 mm.
- **High Modulus (ABS, rPC):**
  - Relies on load distribution due to increased flexural stiffness, distributing forces across a wider EPS liner area.
  - rPC Energy Absorption Ratio: 33.5%, Crush Depth: 10.23 mm.

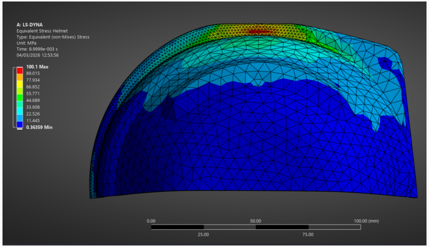
Phase 5: 30° Oblique Impact

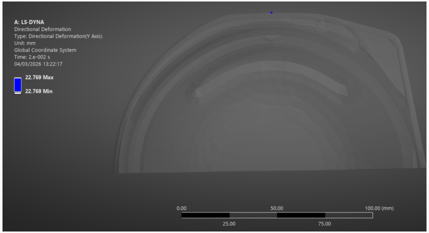
- **Objective:** Advanced simulation beyond standard EN 397 to evaluate real-world complex impacts.
- **Setup:** 30° angled impact surface.
- **Focus:** Rotational kinematics and tangential frictional interactions.



Rotational Kinematics (Key Finding)

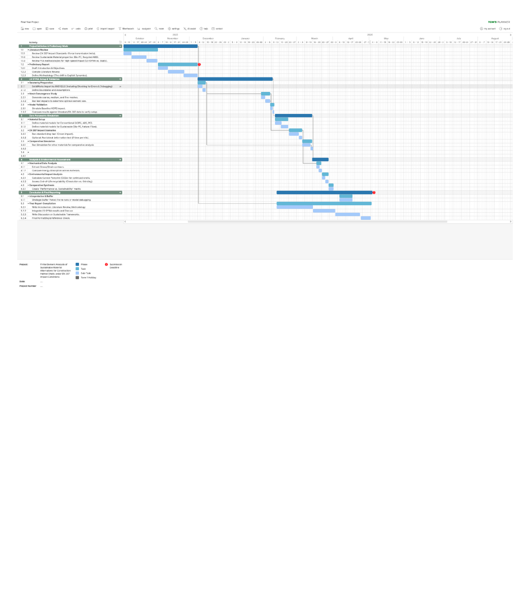
- **rPC Performance:** Achieved a 74% reduction in peak rotational acceleration compared to Virgin HDPE ($896\,\mathrm{rad/s^2}$ vs $3482\,\mathrm{rad/s^2}$).
- **Mechanism:** Reduced tangential frictional interaction. Higher flexural modulus limits localized surface deformation, unlike the plastic deformation seen in HDPE.





Project Management & Risk

- **Execution:** Systematic adherence to the project Gantt chart.
- **Safety:** Implementation of a comprehensive Systematic Risk Register.
- **Audit:** Units synchronized across all models (GPa/kg/mm^2).



Conclusion

- **Validation:** Sustainable alternatives (rPC, Bamboo-HDPE) provide sufficient structural protection and meet strict EN 397 regulations.
- **Superiority in Complex Impacts:** rPC outperforms traditional materials in oblique impacts by mitigating rotational acceleration.
- **Impact:** Significant reduction in the environmental footprint of PPE without compromising safety-critical performance.

Questions?